

Shallow Seismic Velocity Estimation Using Earthquake Particle Motion to Estimate Seismic Hazard Potential

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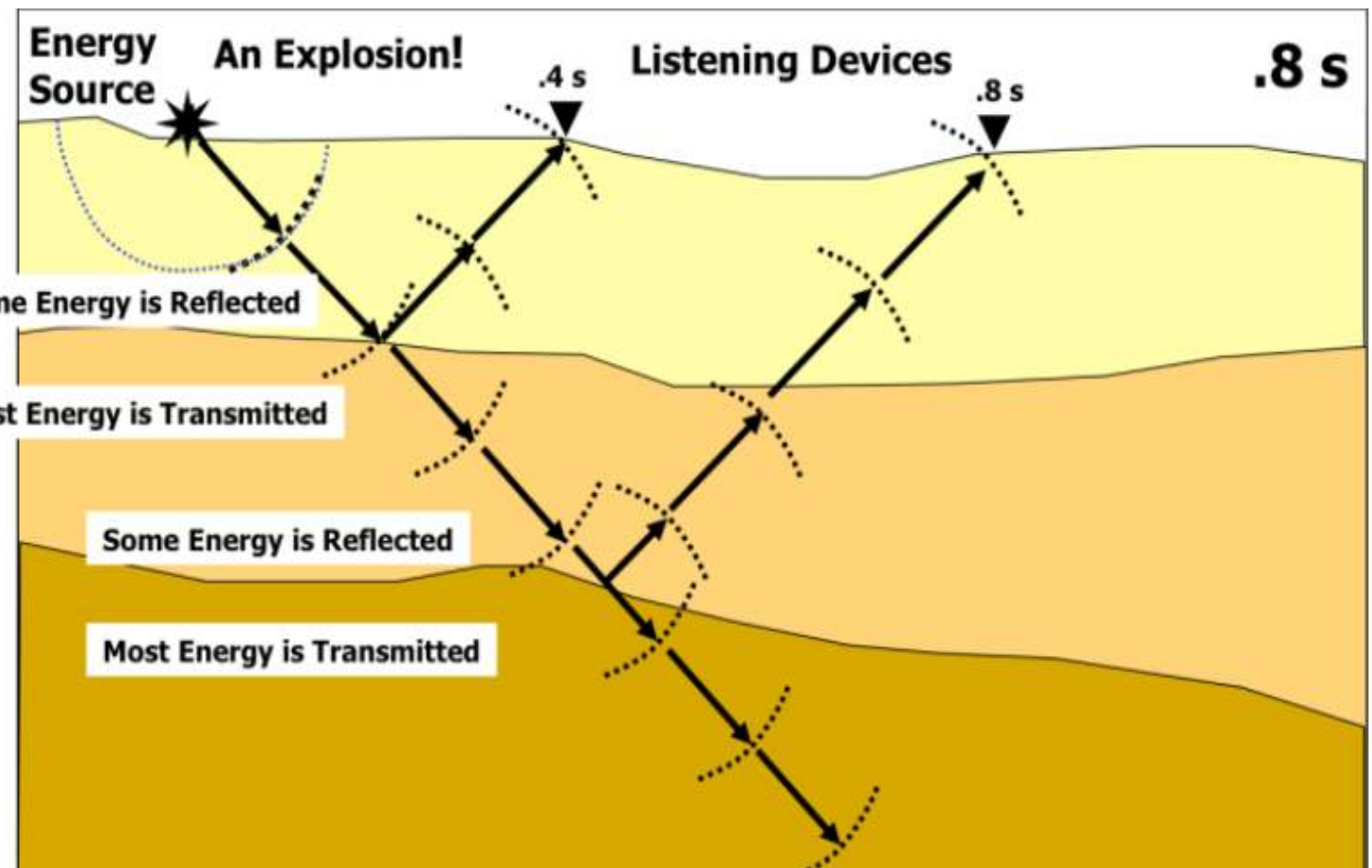
Introduction

How can we use seismic data to better estimate where high shaking will occur in earthquake-prone areas?

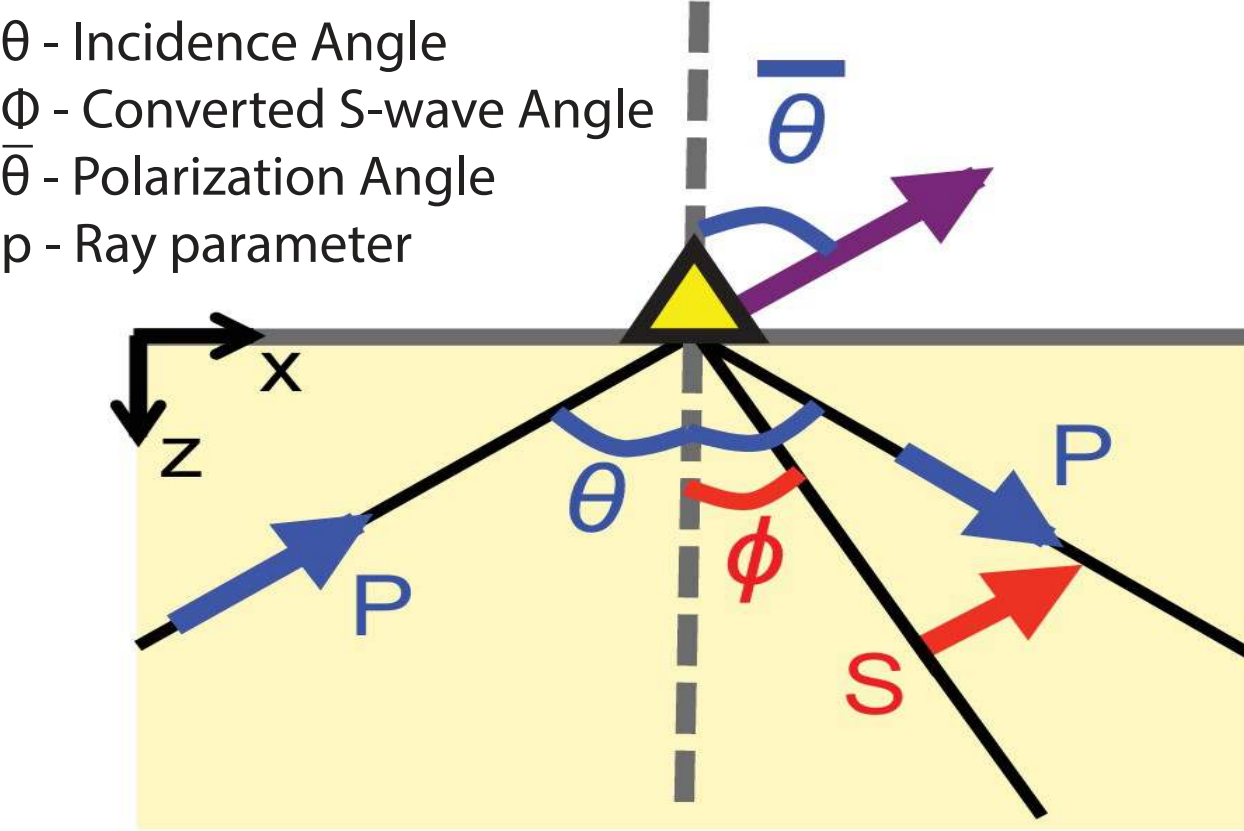
There are various methods currently used to analyze near-surface geologic properties that controls the amount of groundshaking during an earthquake. The approach is to estimate shallow shear-wave velocity structure. We do this through estimating the motion of the Earth due to incoming seismic waves, which are related to the shaking potential.

Shear wave velocities are the fastest in bedrock but, the foundation is strong as well where the infrastructure is built. The shaking is much lesser here. But, in the presence of soft sediments shear wave velocities reduces but the infrastructure experiences more shaking due to a weaker foundation. Hence, it becomes important to image these shallow structures to prevent major damages during earthquakes.

The advantage of these computational methods is that it is non-invasive to the environment, cheaper, and uses naturally occurring seismic waves. More invasive techniques like borehole or explosion analysis relies on artificially created waves that damage the environment. The speeds of these waves are then measured and then an interpretation what lies beneath the surface is made.

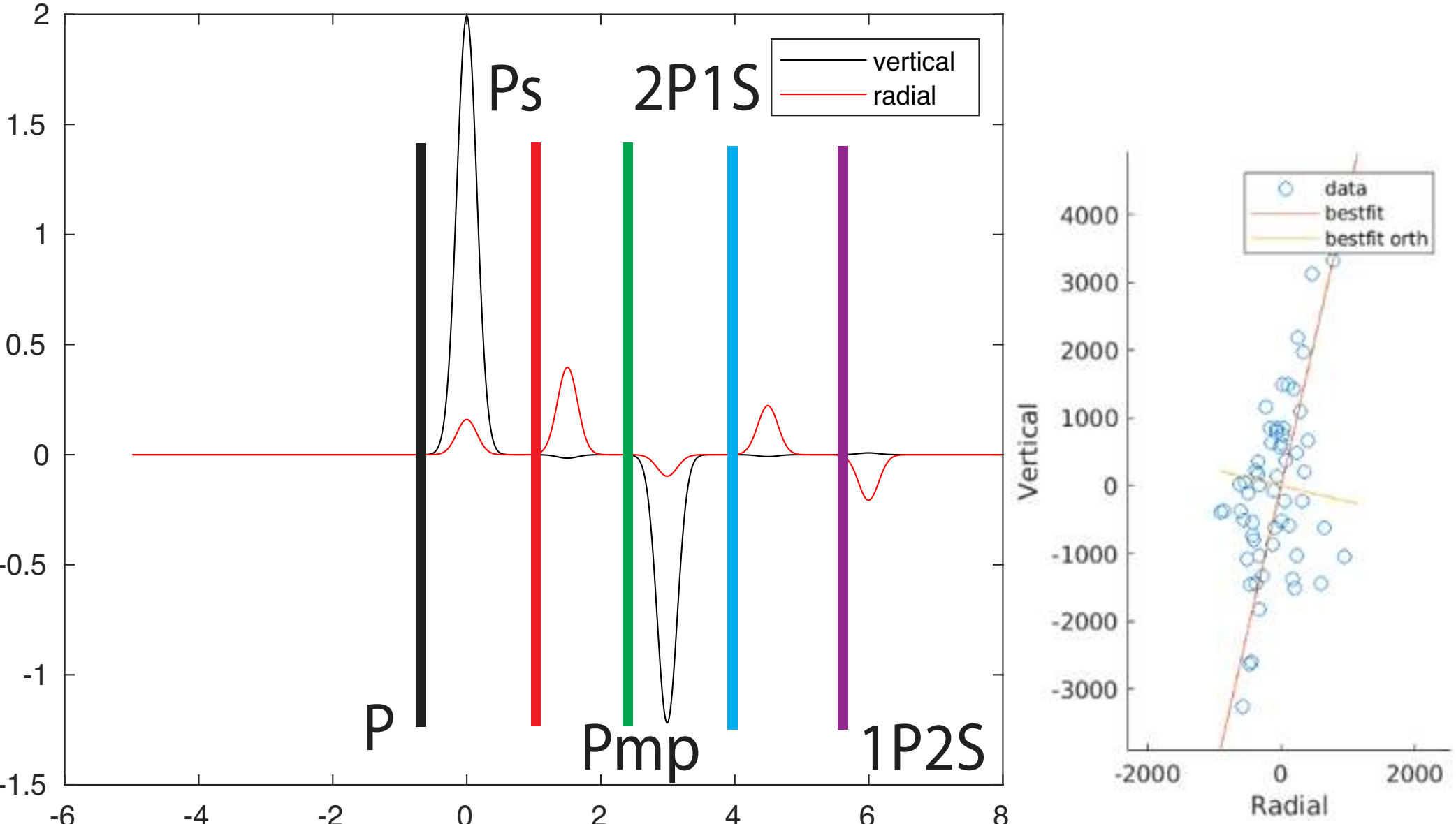
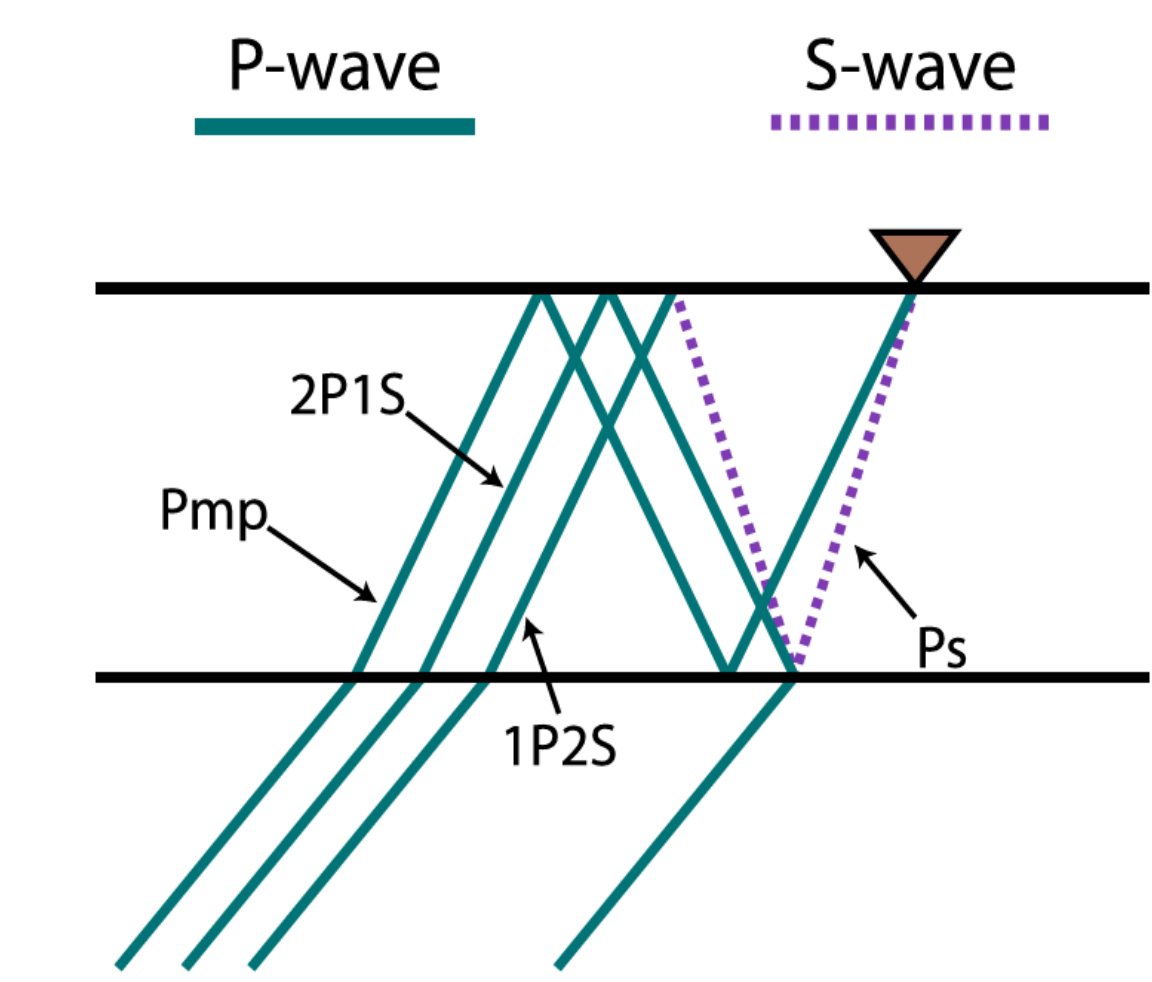


Data and Methods



Particle motion refers to the movement of particles as seismic waves pass through. This motion is three-dimensional and can be represented by a vector, which has both direction and magnitude. P-waves causes the particles to move back-forth, while S-wave move particles perpendicular to the wave propagation. Seismic instruments, measure this ground motion caused by seismic waves. An angle of the motion can be inferred from the seismogram which can be used to measure speeds.

$$\beta = \frac{\sin \phi}{p} = \frac{\sin \left(\frac{\bar{\theta}}{2} \right)}{p}$$

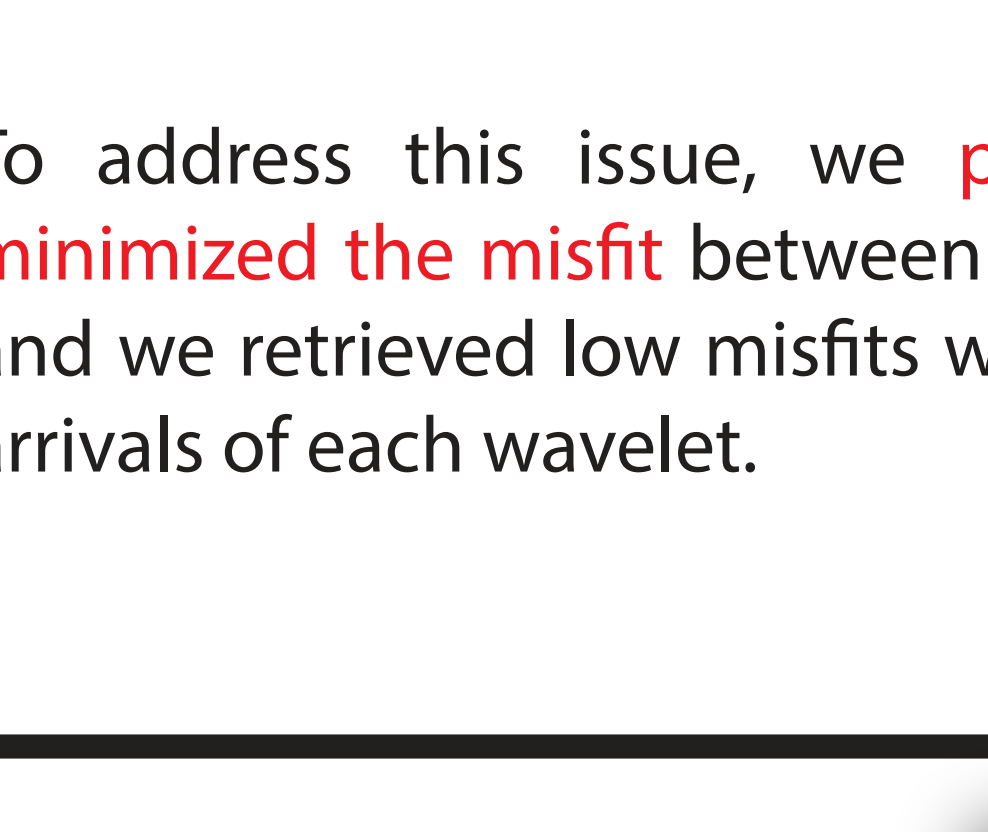
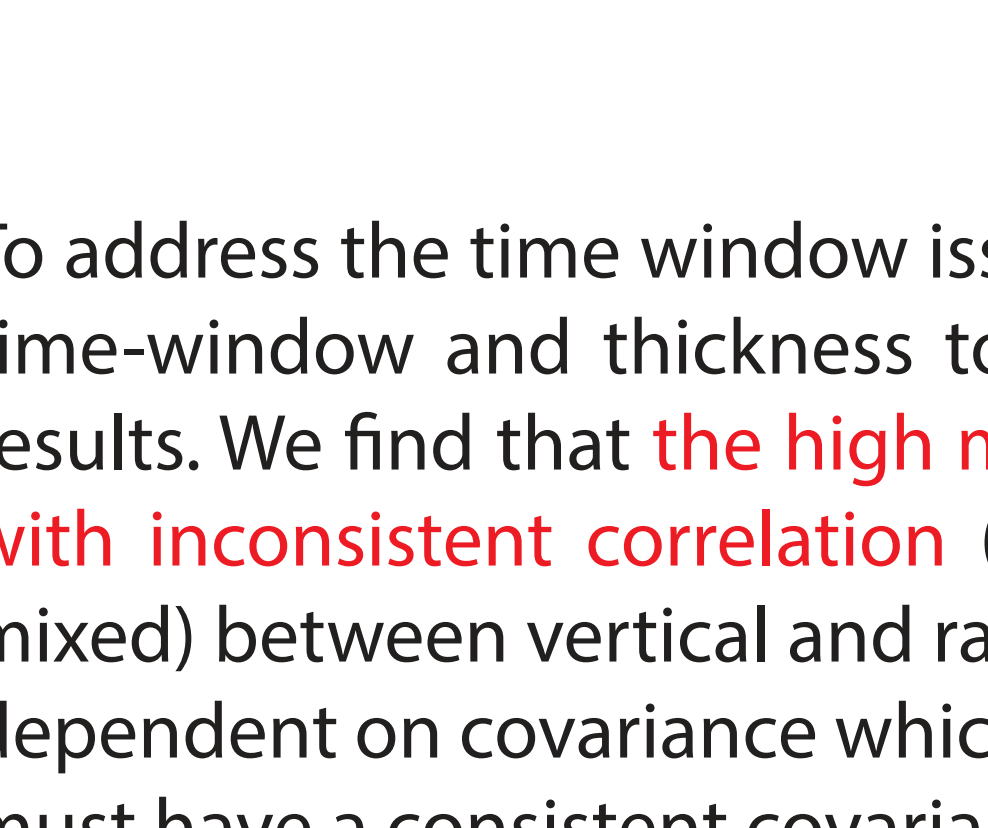
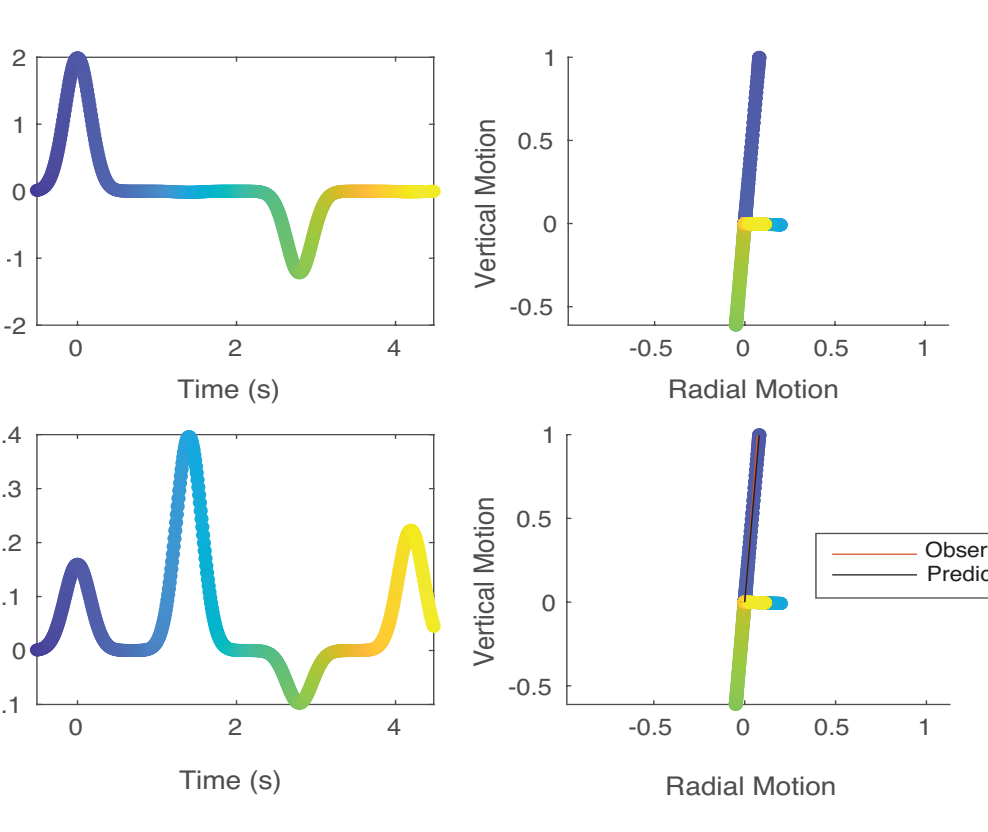
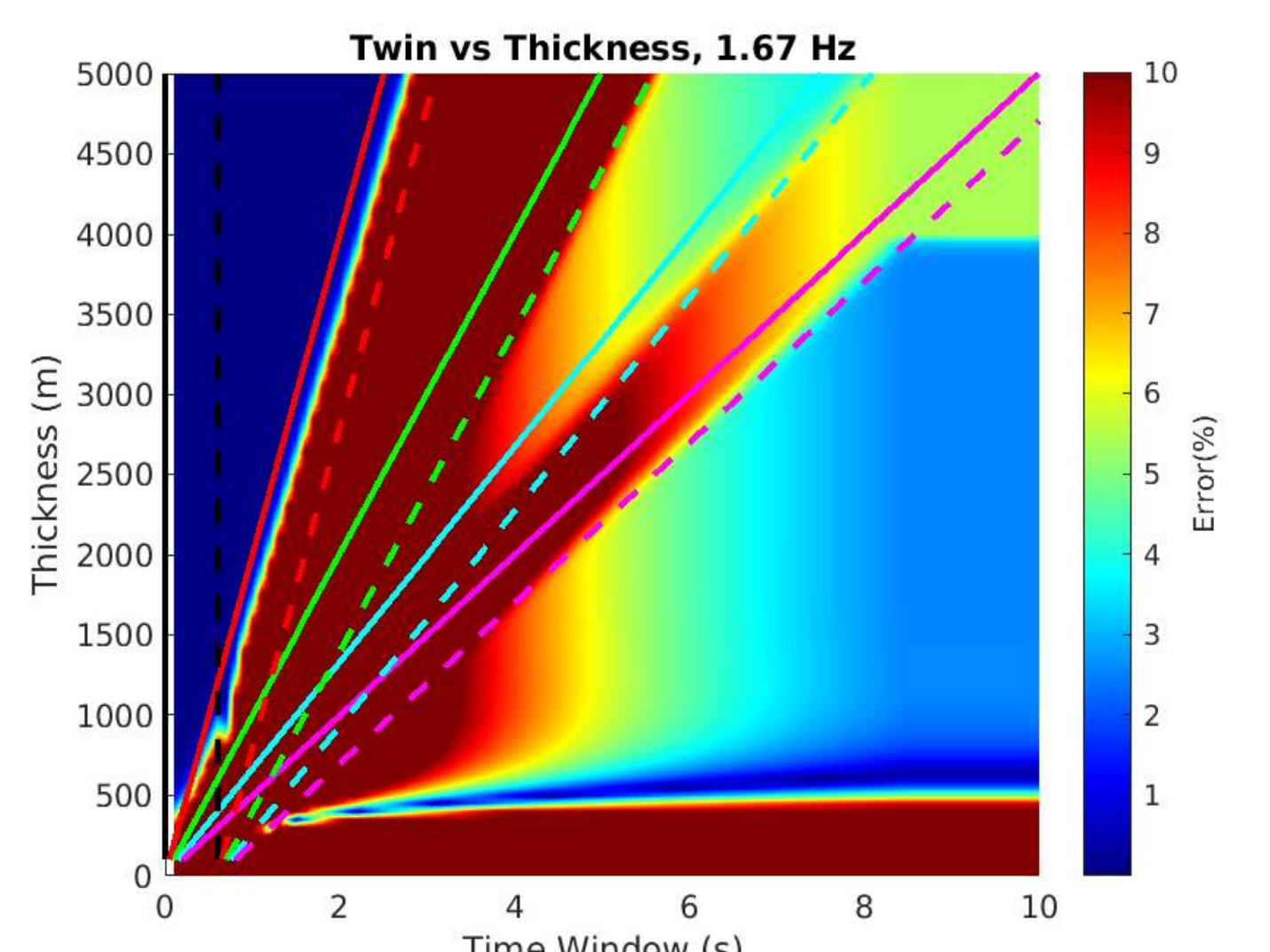
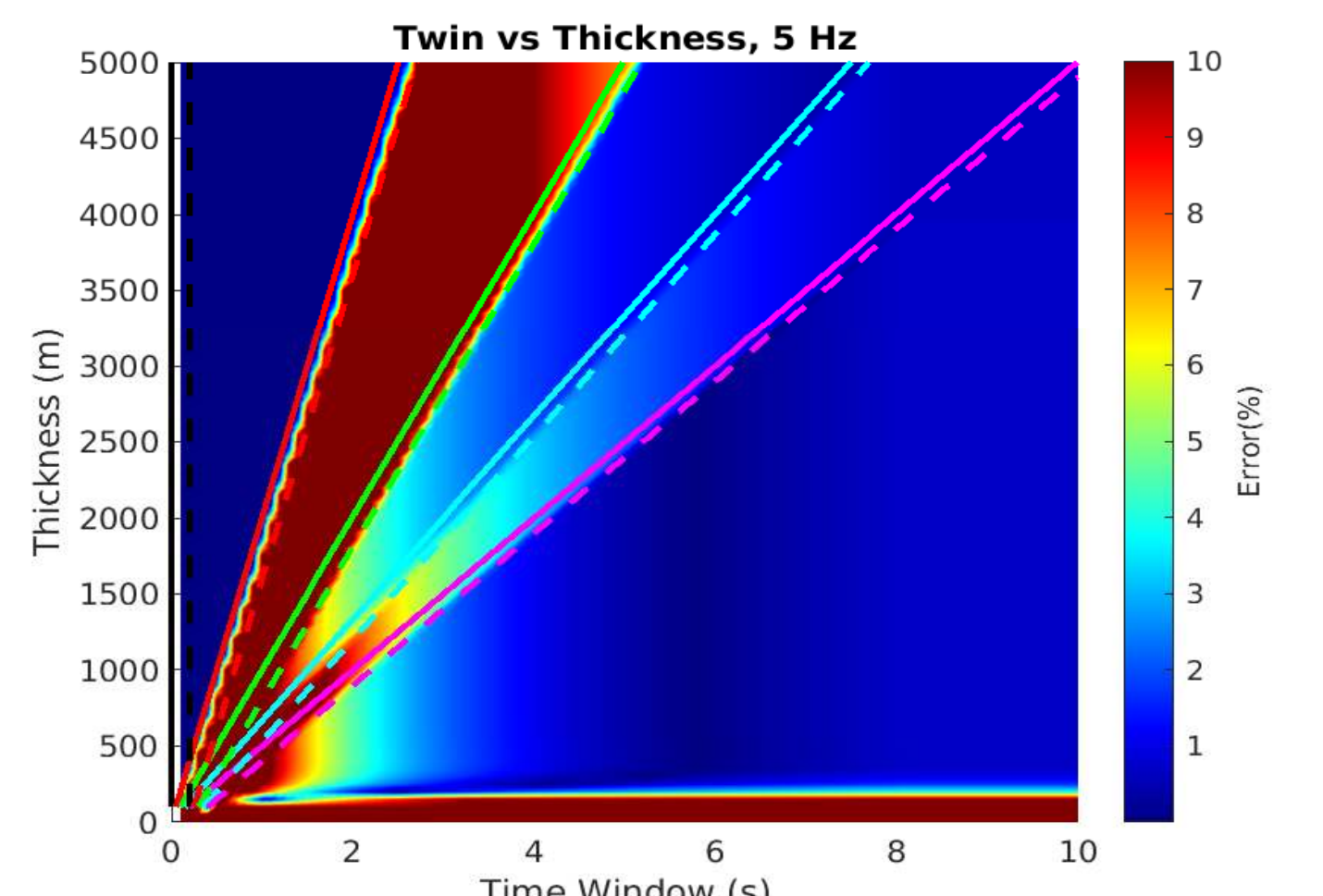
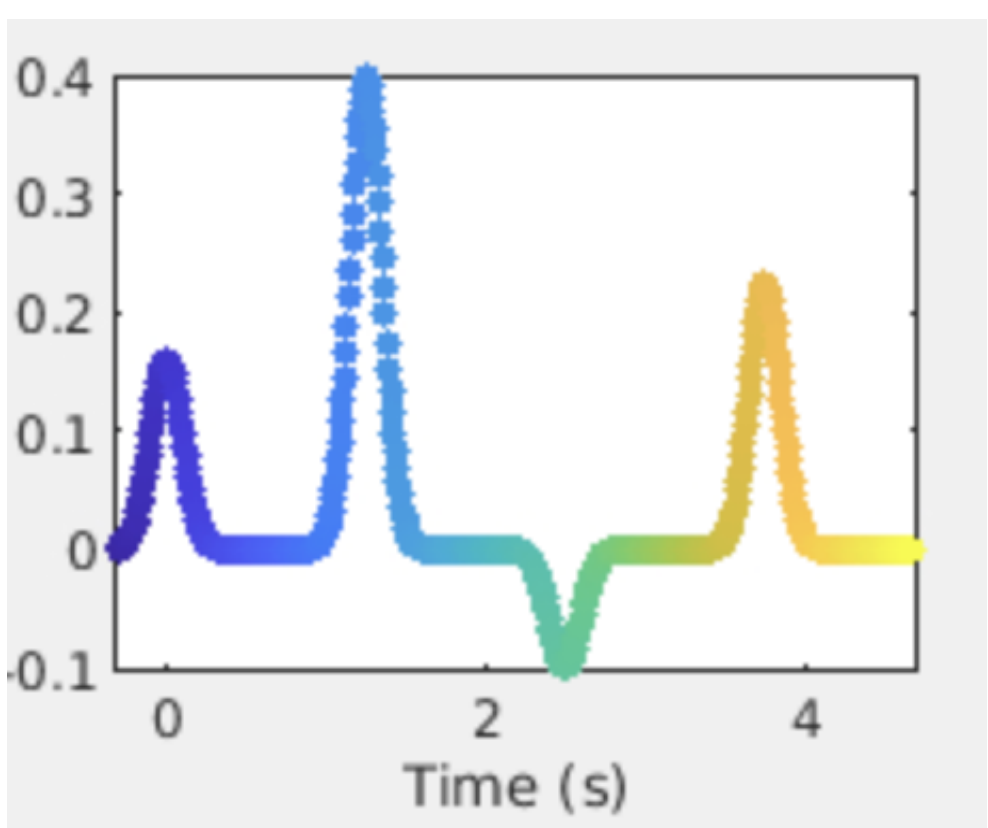
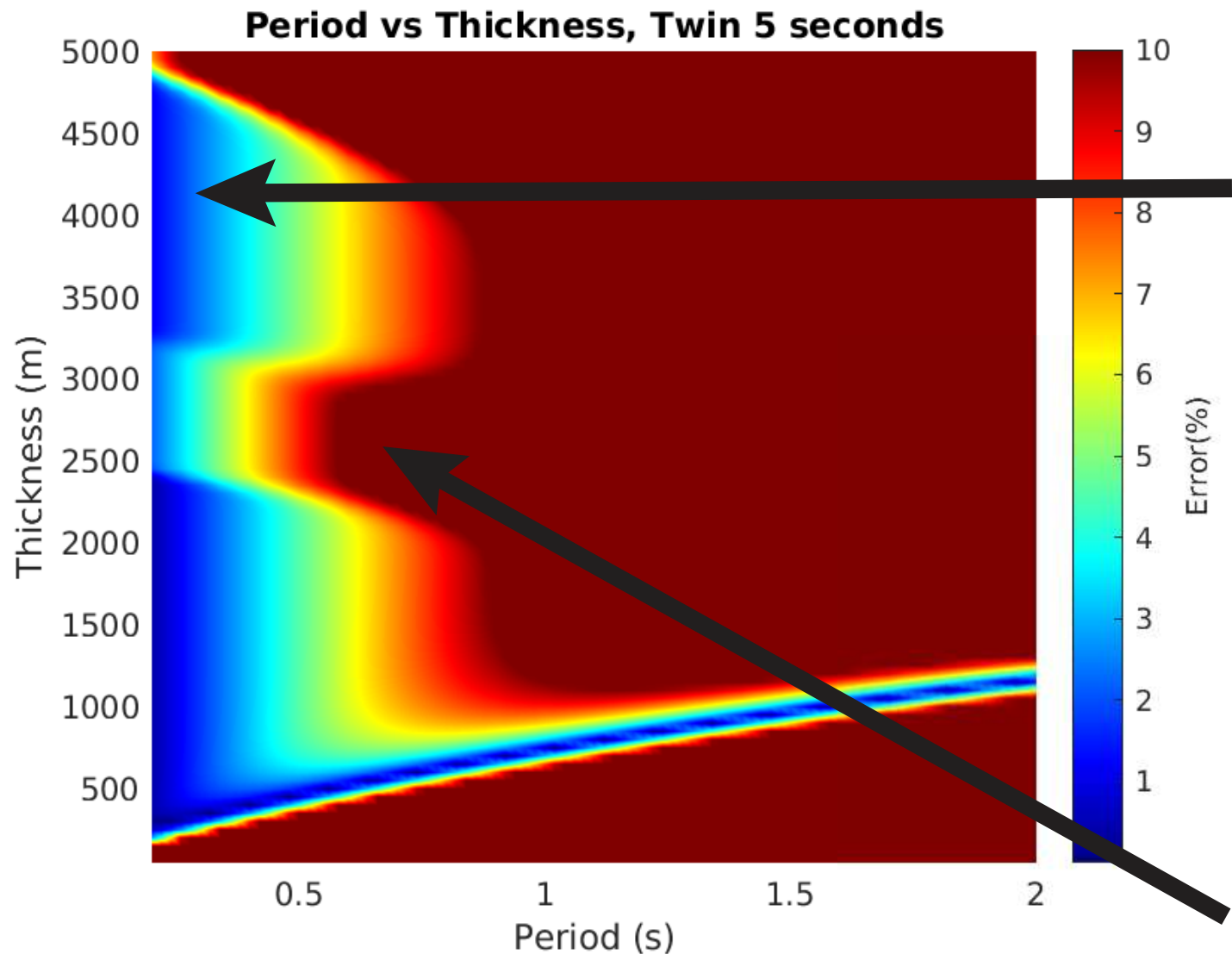


References

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Synthetic Results

PCA



We created a synthetic dataset over different frequency contents, layer thickness, and time-window for particle motion analysis to understand over what parameter space combinations we should recover accurate results.

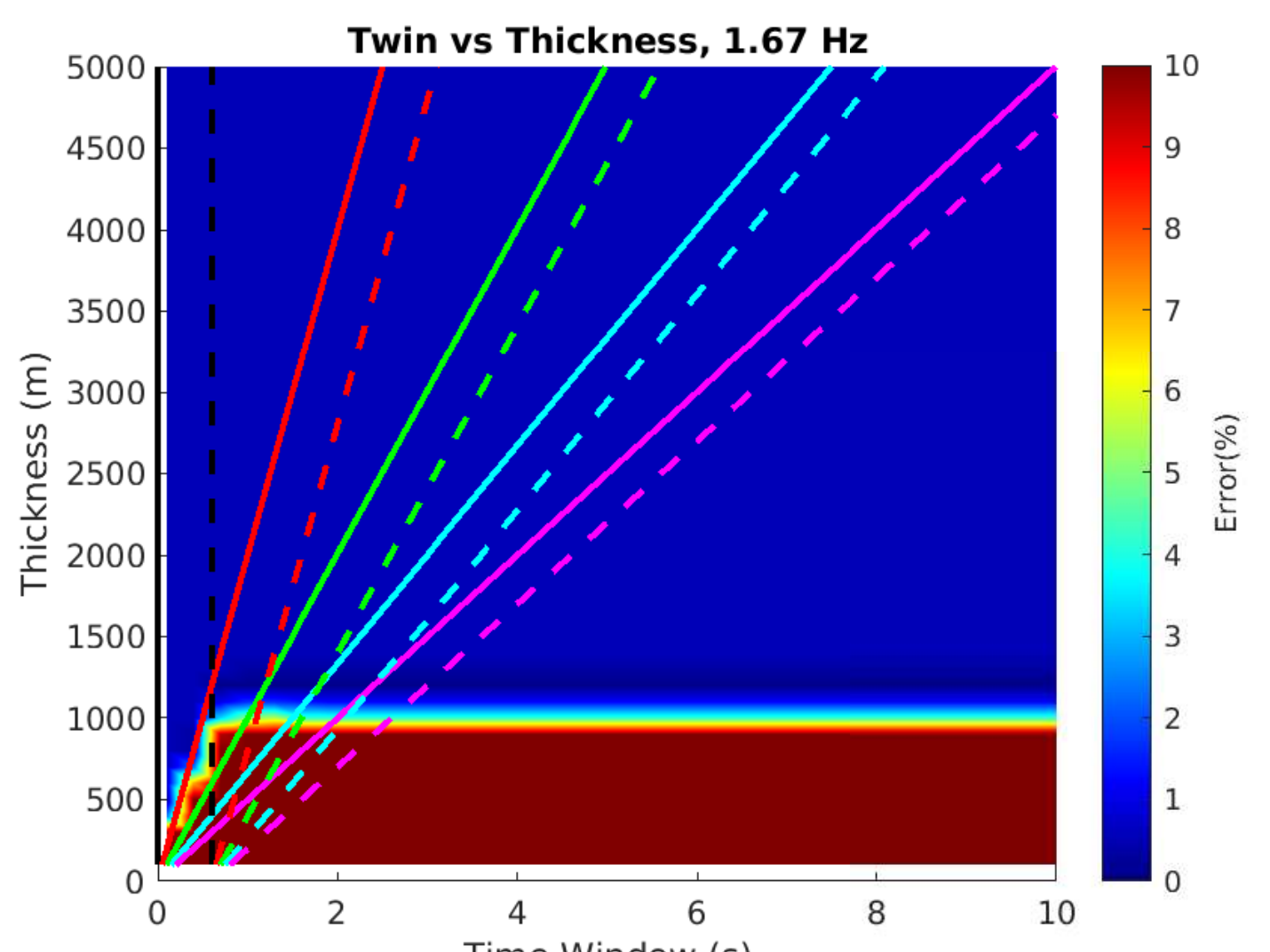
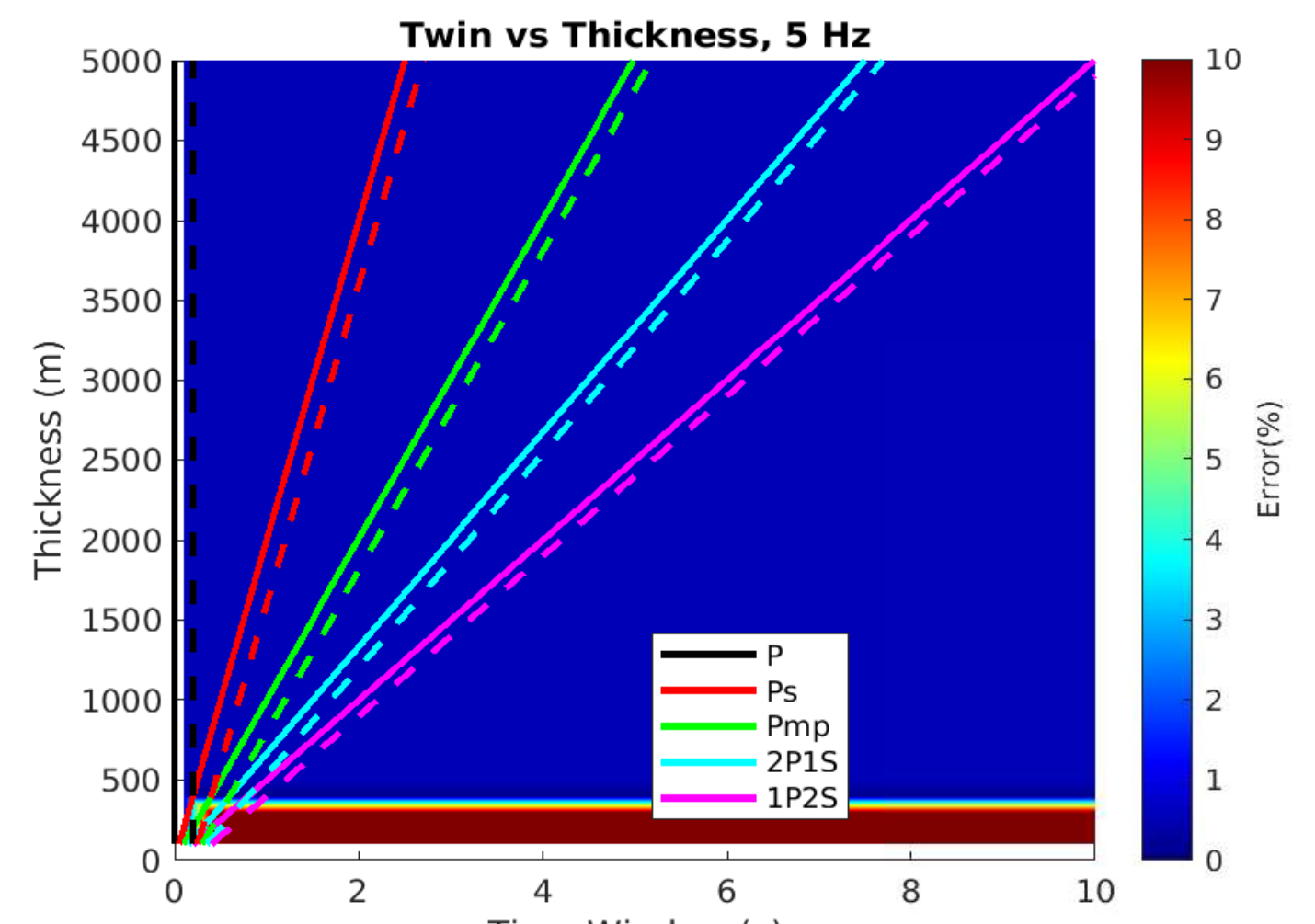
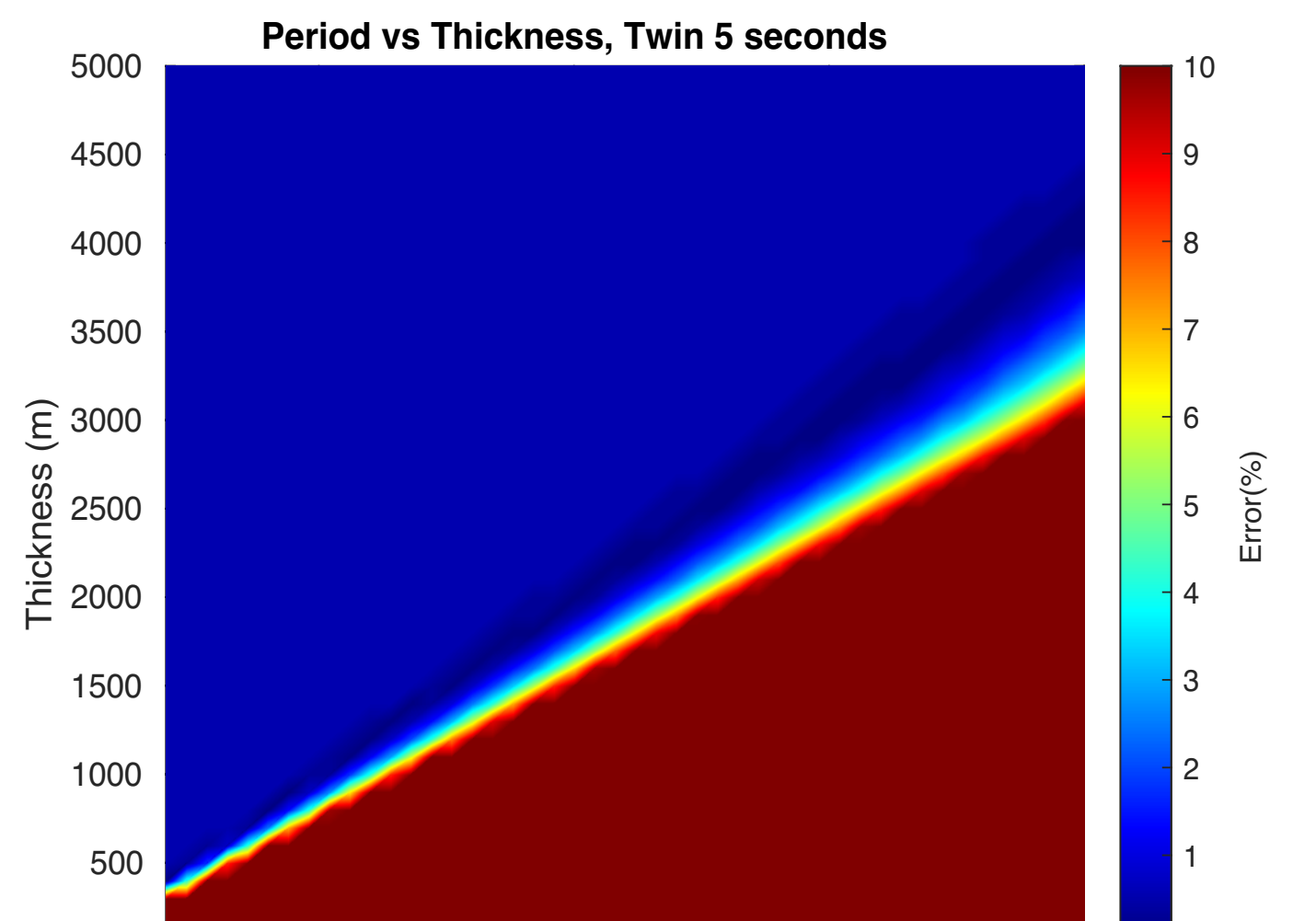
We used principal component analysis (PCA) to estimate the orthogonal vectors that characterize the particle motion of our modelled earthquakes.

We first fixed time window at 5 seconds, and found a “dip” in sensitivity between [2500,3000] meters that deviates from theoretical expectations. This happened because of the amplitude of last wavelet didn’t reach 0 leading to a bias in the PCA result. Expanding the time window cannot address this due to shortcomings in the PCA algorithm.

To address the time window issue, we fixed frequency and searched over time-window and thickness to find out where we can obtain accurate results. We find that the high misfit occurs at arrivals of certain wavelets with inconsistent correlation (i.e., not always positive or negative, but mixed) between vertical and radial components. This is because PCA is dependent on covariance which simply means that the vertical and radial must have a consistent covariance for the algorithm to work.

To address this issue, we performed grid-search over angles that minimized the misfit between orthogonal lines and our particle motion and we retrieved low misfits which remained consistent throughout the arrivals of each wavelet.

GRID



SCENTAR Results

We applied PCA and grid-search analysis on seismic data recorded by 60 stations in Northern California. A total of 20 seismic events were recorded, and after a quality check we used 16 of those events which had a clear seismic signal. The shear wave velocities in certain regions were as low as 1.8 km/s which indicates potential of high earthquake shaking. Higher shear wave velocities could mean presence of consolidated rocks (i.e., low shaking). We also made a plot comparing the grid-search to PCA velocities and find that on average the grid-search provided lower velocities than PCA.

